a score denoting how helpful that point is in helping us identify the global optimum. This score is sometimes called the *acquisition score*. Our job is then to pick out the point that maximizes this acquisition score and evaluate the objective function at that point.

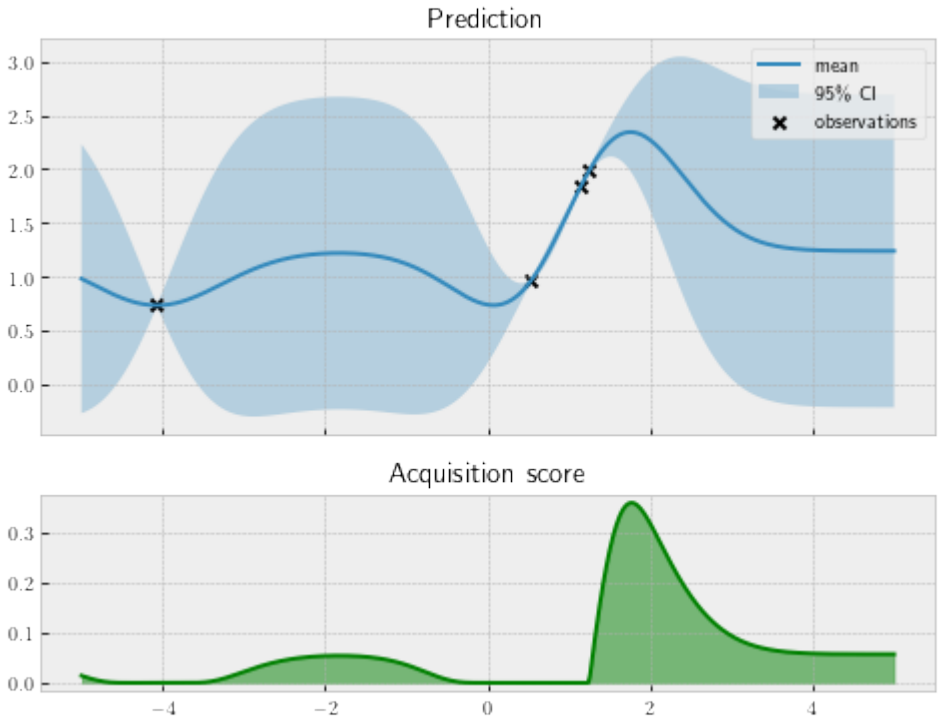


Figure 1.5 A Bayesian optimization policy scores each individual data point by their usefulness in locating the global optimum. The policy prefers high predictive values (where the payoff is more likely) as well as high uncertainty (where the payoff may be large).

Figure 1.5 shows the same GP that we have in Figure 1.4, where the bottom panel shows the plot of how a particular Bayesian optimization policy called *Expected Improvement* scores each point on the *x*-axis between -5 and 5 (which is our search space). We will learn about what this name means and how the policy scores data points in Chapter 4. For now, let's just keep in mind that if a point has a large acquisition score, it means that this point is valuable in locating the global optimum.

In Figure 1.5, the best point is around 1.8, which makes sense as according to our GP in the top bottom, that's also where we achieve the highest predictive mean. This means we will then pick this point at 1.8 to evaluate our objective, hoping to improve from the highest value we have collected.

We should note that this is not a one-time procedure but instead a *learning loop*. At each iteration of the loop, we train a GP on the data we have observed from the objective, run a Bayesian optimization policy on this GP to obtain a recommendation that will hopefully help us

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